ModelIt!

Teacher's Guide

Does More Beavers Mean a Bigger Dam?

How Beaver Builders Shape Their World

Quick Reference

Grade Level	Grade NE
NGSS Standard	3-LS4-3, 3-LS4-4, 3-5-ETS1-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 3-LS4-3, 3-LS4-4, 3-5-ETS1-1

Construct an argument that in a particular habitat some organisms can survive well, some survive less well, and some cannot survive at all. Make a claim about the merit of a solution to a problem caused when the environment changes. Define a simple design problem reflecting a need or want that includes criteria for success.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a computational model showing how beaver population and tree availability affect dam and pond size.
Disciplinary Core Idea	**LS4.C Adaptation** — Beavers are adapted to build dams, which changes the environment to better suit their survival needs and creates habitat for other species.
Crosscutting Concept	**Cause and Effect** — Students identify that changes in beaver population or tree availability cause predictable effects on dam size and pond size.

CAST Testing Alignment

MULTIPLE CHOICE:

A beaver family moves to a stream with lots of trees. Over time, what will MOST LIKELY happen? A) The stream stays the same B) A pond forms behind a new dam C) The trees grow taller D) The beavers move away

CONSTRUCTED RESPONSE:

Using what you learned from your model, explain what would happen if a beaver family tried to build a dam but there were almost no trees nearby. Use the words 'dam size' and 'pond size' in your answer.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

Beavers are one of nature's most impressive builders. They are the only animals besides humans that significantly change their environment by building structures. Understanding how beavers shape ecosystems helps scientists manage wetlands and watersheds.

Beavers as Ecosystem Engineers

An ecosystem engineer is an organism that creates, significantly modifies, or destroys a habitat. Beavers are the classic example. By felling trees and building dams, they transform flowing streams into still ponds, creating entirely new ecosystems. A single beaver family can transform acres of landscape.

How Beaver Dams Work

Beavers use their large front teeth to cut down trees, then drag branches and logs to a stream. They pack mud and vegetation between the logs to make the dam watertight. The dam slows the stream and creates a pond. Beaver dams can be tiny or massive — the longest known beaver dam in Alberta, Canada is over 2,700 feet long!

Why Dams Matter

The ponds created by beaver dams provide habitat for fish, amphibians, birds, insects, and plants. They also filter water, reduce flooding downstream, and recharge groundwater. When beavers are removed from an area, the dam eventually breaks down, the pond drains, and all the species that depended on it must find new homes.

Beavers and Trees

Beavers prefer softwood trees like willow, aspen, and cottonwood. They eat the bark and use the wood for building. A single beaver can fell a tree 6 inches in diameter in about 15 minutes. If beavers use up all nearby trees, they must either travel farther (risking predators) or move to a new location.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify Number of Beavers and Trees Available as external components (things we can set) and Dam Size and Pond Size as internal components (things that change based on the externals).

E **Establish Relationships**
Students determine that both Number of Beavers and Trees Available positively affect Dam Size, and Dam Size positively affects Pond Size — but BOTH externals are needed.

V **Visualize & Model**
Students build a model showing how two external inputs feed into Dam Size, which then feeds into Pond Size.

E **Evaluate Outcomes**
Students run scenarios with different beaver/tree combinations and observe that removing either input collapses the system.

R **Revise & Extend**
Students add Rainfall or Number of Other Animals to see how the pond creates a ripple effect in the ecosystem.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Have you ever seen a beaver? They are one of nature's greatest builders! Today we are going to figure out HOW they build and WHY it matters."*

Watch For

Show a short video clip of beavers building a dam (30 seconds). Ask: What do you notice?

Slide 2: Learning Objectives

3 min

Say: *"Today we are going to build our own computer model to figure out what makes a beaver dam successful."*

Watch For

Have students read objectives aloud. Pre-teach 'ecosystem engineer' with examples.

Slide 3: Big Question

3 min

Say: *"Here is our big question. What do YOU think happens when beavers start building?"*

Watch For

Have students turn and talk: What do you think a beaver needs to build a dam? Share out.

Slide 4: LEVER Framework

2 min

Say: *"We are going to use a step-by-step process called LEVER to build our model. Each letter stands for a step."*

Watch For

Briefly introduce each LEVER step. Explain that models help us test ideas without going to a real beaver pond.

Slide 5: Activity 1: Components

8 min

Say: *"Let us figure out the parts of our system. What things affect the dam? What things CHANGE because of the dam?"*

Watch For

Guide component sorting: Number of Beavers and Trees Available are things WE set (external). Dam Size and Pond Size CHANGE on their own (internal).

Slide 6: Activity 2: Connections

8 min

Say: *"Now let us connect them! When the number of beavers goes UP, does dam size go UP or DOWN?"*

Watch For

Help students draw positive arrows. Key moment: What happens when trees go to zero? Dam size drops no matter how many beavers there are.

Slide 7: Activity 3: Simulation

10 min

Say: *"Time to test! First, PREDICT what will happen. Then we will run the model and see if you were right."*

Watch For

Run all three scenarios. Have students record predictions BEFORE running each one. Compare results to predictions.

Slide 8: Discoveries

5 min

Say: *"So what did we discover? Beavers are amazing builders — but they need their team AND their materials!"*

Watch For

Summarize key findings. Ask: What other animals change their environment?

Slide 9: STEM Challenge

12 min

Say: *"Now YOU are the beaver engineers! Build a dam that holds back the most water."*

Watch For

Distribute materials (craft sticks, clay, plastic bins). Groups design and test dams. Measure water held.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Number of Beavers, Trees Available
- **INTERNAL:** Dam Size, Pond Size

Relationships

- **Number of Beavers to Dam Size:** POSITIVE (+) — More beavers means more workers to cut trees, haul branches, and pack mud. A larger beaver family can build a bigger dam faster.
- **Trees Available to Dam Size:** POSITIVE (+) — More trees means more building material. Even a large beaver family cannot build much without trees nearby.
- **Dam Size to Pond Size:** POSITIVE (+) — A bigger dam holds back more water, creating a larger and deeper pond behind it.

Key Discoveries

- Beavers need BOTH enough family members AND enough trees to build a big dam
- A bigger dam holds back more water, creating a bigger pond
- Even lots of beavers cannot build without trees — both inputs matter
- Beavers are ecosystem engineers because their building changes the whole environment

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Beavers build dams to live inside"	Beavers actually live in a separate structure called a lodge — a dome-shaped pile of sticks in the middle of the pond. The dam creates the pond, and the lodge is their home IN the pond. The dam and the lodge are two different structures!
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Dams are bad for the environment"	Human-made dams can sometimes cause problems, but BEAVER dams are part of the natural ecosystem. They create wetlands that support hundreds of species, filter water, reduce flooding, and recharge groundwater. Beaver dams are one of nature's best tools for keeping water systems healthy!
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"One beaver can build a big dam by itself"	While a single beaver CAN start a dam, building and maintaining a large dam requires a family group (colony) of 4-8 beavers. Dam building is teamwork! A big dam needs constant repair, and one beaver cannot keep up with the work alone.